

ST JOSEPH, MI, USA

WMO: 998012

Lat: 42.0980N

Lon: 86.4940W

Elev: 604

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.2	(c) 10.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 32.0	(k) 27.5	(l) 29.1	(m) 26.4	(n) 9.1	(o) 160	(p) 0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.2	(c) 85.9	(d) N/A	(e) 83.0	(f) N/A	(g) 80.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 28.4	(b) 24.9	(c) 21.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	25.8	27.4	36.6	48.5	58.8	67.4	72.7	71.2	65.1	54.1	42.5	34.0
	DBStd	18.08	10.13	9.32	9.50	8.01	8.54	6.41	5.83	4.76	6.62	7.14	8.69	8.38
	HDD50	2750	750	633	436	122	17	0	0	0	0	36	254	502
	HDD65	6007	1215	1052	883	498	234	48	9	7	79	346	674	962
	CDD50	2921	1	0	19	78	289	523	704	656	453	163	30	5
	CDD65	702	0	0	1	4	40	121	247	199	82	8	0	0
	CDH74	3464	0	0	3	25	208	568	1430	874	332	24	0	0
	CDH80	632	0	0	0	1	42	99	312	128	48	2	0	0
Wind	WSAvg	8.0	11.0	9.3	8.1	8.1	6.2	5.8	6.0	5.7	6.2	8.3	10.6	10.3
Precipitation	PrecAvg	35.2	2.3	1.8	2.0	3.0	3.4	3.5	3.3	4.0	3.5	3.6	3.1	2.3
	PrecMax	45.2	4.9	4.3	4.2	5.2	6.7	7.6	7.2	10.9	11.0	8.7	7.2	5.7
	PrecMin	27.4	0.6	0.5	0.5	0.3	0.7	1.0	0.8	1.5	0.9	0.5	1.1	1.0
	PrecStd	5.1	1.1	0.9	1.0	1.3	1.7	1.9	1.5	2.2	2.3	2.1	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	56.8	73.7	77.7	85.1	86.8	90.1	87.3	85.1	77.4	67.5	61.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	50.8	50.1	65.7	72.4	79.3	82.9	86.6	83.4	81.0	71.5	62.6	55.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	43.4	43.8	57.7	66.7	75.4	80.3	83.4	80.7	78.0	68.7	58.9	49.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	39.2	38.9	50.5	61.9	71.4	77.6	80.8	78.5	75.1	65.4	55.1	45.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.4	12.1	14.6	14.9	13.2	12.2	12.5	13.4	12.4	11.8	9.9
		MCDBR	14.4	17.9	20.9	23.5	19.2	17.3	15.6	15.5	15.8	16.4	16.3	15.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.304	0.326	0.346	0.392	0.421	0.432	0.439	0.436	0.409	0.358	0.322	0.305
	taud		2.433	2.359	2.371	2.277	2.257	2.278	2.258	2.266	2.304	2.444	2.492	2.475
	Ebn at Noon		267	277	285	278	272	268	265	262	260	263	259	257
	Edh at Noon		24	31	34	41	43	42	42	41	36	28	23	22
All-Sky Solar Radiation	RadAvg		419	672	1127	1504	1726	1968	1997	1715	1385	881	524	352
	RadStd		44	90	90	128	154	126	93	99	116	98	80	44

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page